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Aryabhatta Knowledge University, Patna

B.Tech 4th Semester Examination, 2022

Time: 03 Hours

Code: 110405 ***Subject: Microprocessors*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option of the following (any seven) : **[2x7=14]**

(a) Intel 8085 is a
- (i) microprocessor
- (ii) microcontroller
- (iii) Both (i) and (ii)
- (iv) None of the above

(b) On reset of 8051, stack pointer points to _____ address.
- (i) 00H
- (ii) 05H
- (iii) 07H
- (iv) 0AH

(c) When 8051 wakes up, then 0x00 is loaded to which register?
- (i) DPTR
- (ii) SP
- (iii) PC
- (iv) PSW

(d) The register whose bits represent status flags is
- (i) DPTR
- (ii) SP
- (iii) PC
- (iv) PSW

(e) MOVC is used in _____ addressing mode.
- (i) direct
- (ii) indirect
- (iii) index
- (iv) None of the above

(f) Device pins XTAL1 and XTAL2 of 8051 are used for connections to
- (i) external oscillator
- (ii) external power
- (iii) external peripheral
- (iv) external memory

(g) Which of the 8051 variant architectures does not have ROM?
- (i) 8051
- (ii) 8052
- (iii) 8031
- (iv) None of the above

(h) SWAP instruction works on _____ register.
- (i) any
- (ii) accumulator
- (iii) general purpose
- (iv) special function

(i) 8051 can handle _____ interrupt sources.
- (i) 3
- (ii) 4
- (iii) 5
- (iv) 6

(j) Name the architecture and the instruction set for microcontroller.
- (i) von-Neumann architecture with CISC instruction set
- (ii) Harvard architecture with CISC instruction set
- (iii) von-Neumann architecture with RISC instruction set
- (iv) Harvard architecure with RISC instruction set

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Q.2 Solve all three questions : **[6+5+3=14]**

(a) Differentiate between Microprocessor and Microcontroller. Also list some of the popular microprocessors and microcontrollers.

(b) What is word size of a processor? How is this related to ALU size, Memory size, Register size and Bus size? Explain.

(c) Comment on the benefit of having separate program and data memory in the typical microcontroller architecture.

Q.3 Solve both questions : **[7+7=14]**

(a) What make any system embedded? Give examples of application areas of embedded systems. Write three key characteristics of an embedded system.

(b) Discuss about the role of micro-controllers in embedded systems in detail. Give a detailed note on the technologies other than microcontrollers used in embedded system design.

Q.4 Solve all three questions : **[6+4+4=14]**

(a) List addressing modes of 8051. By how many different ways we can access the program/code memory? Explain by using suitable example instructions.

(b) Write secondary functions of Port P3 pins of 8051.

(c) Differentiate among MOV, MOVC and MOVX instructions.

Q.5 Solve both questions : **[7+7=14]**

(a) Comment on the ROM availability in different variant architectures of 8051. How to access an external ROM? Draw proper interfacing diagram.

(b) How is machine cycle related to clock frequency and the execution time of an instruction? Explain in detail.

Q.6 Solve both questions : **[7+7=14]**

(a) Write 8051 assembly program to copy a block of 10 bytes from RAM location starting at 37h to RAM location starting at 59h.

(b) Explain the role of timers in 8051. Also explain the registers used for timer operations.

Q.7 Solve both questions : **[6+8=14]**

(a) Write a program to generate a delay of 10 milliseconds using timer.

(b) How can we use time delay generated in question no. 7(a) for making Diwali lighting with blinking period of 10 milliseconds? Use delay code as subroutine.

Q.8 Solve all three questions : **[6+5+3=14]**

(a) What is ADC and DAC? Explain DAC working with suitable diagram.

(b) What is interrupt? What is need of interrupts in the processor? List 8051 interrupts.

(c) List the serial communication interface standards. Write a few lines on each.

Q.9 Solve all three questions : **[5+4+5=14]**

(a) Differentiate between Synchronous and Asynchronous communications.

(b) What are sensors and actuators? List different sensors and actuators used in embedded system design.

(c) Discuss about the wireless protocols used in communication for embedded systems.

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